

RC discharge theory notes

Synthetic teaching notes authored for ResearchHub.

Model: $V(t) = V_0 * \exp(-t / \tau)$, where $\tau = R * C$.

At one time constant the voltage falls to about 36.8 percent of V_0 .

Expected τ from the nominal component values is 4.7 seconds.

Fit $\log(V)$ against time; a negative slope gives $\tau = -1 / \text{slope}$.

Only positive, finite voltages and finite, nonnegative times are valid.

A log fit assumes negligible offset and roughly multiplicative errors.

Report this assumption: additive voltage noise can bias a log fit.

Component tolerances and measurement noise explain a modest deviation.

Compare measured and fitted curves, and inspect residuals before conclusions.